Calculus with Parametric and Polar Curves

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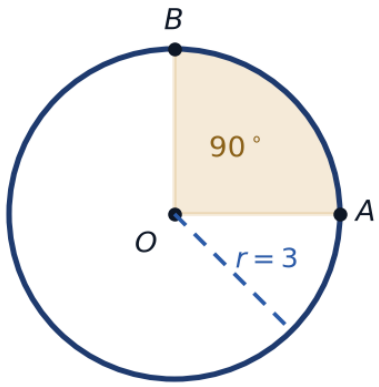
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Parametric derivatives and arc length

For the curve $x = t^2$, $y = t^3$, find $\frac{dy}{dx}$ in terms of t.

2 [Foundational] • Target: Parametric derivatives and arc length

Find the arc length of the curve $x = 3t^2$, $y = 2t^3$ for $0 \le t \le 2$.



3 [Foundational] • Target: Polar area

Find the area enclosed by one loop of the rose $r = 3\sin 2\theta$ (for $0 \le \theta \le \pi/2$).

4 [Foundational] • Target: Polar area

Find the area enclosed by the cardioid $r = 2 + 2\cos\theta$.

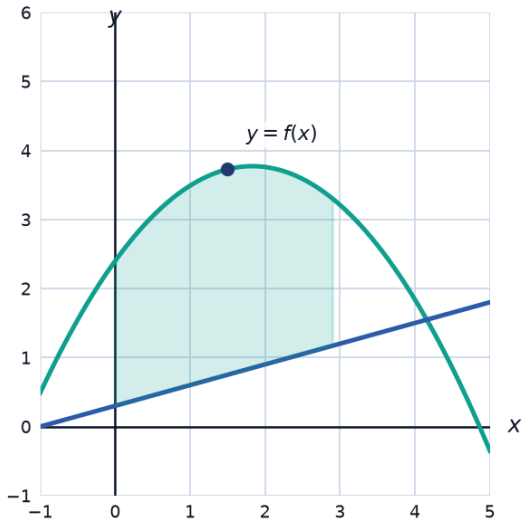
5 [Foundational] • Target: Second derivatives and concavity for parametric curves

For $x = t^2$, $y = t^3 - 3t$, find $\frac{d^2y}{dx^2}$ in terms of t, using $\frac{d^2y}{dx^2} = \left[\frac{d}{dt}\left(\frac{dy}{dx}\right)\right] \div \frac{dx}{dt}$.

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6 [Foundational] • Target: Area between polar curves

Find the area inside the circle $r = 3$ and outside the circle $r = 2$ (an annular region).



7 [Foundational] • Target: Horizontal and vertical tangents on parametric curves

For $x = t^2 - 4$, $y = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal ($\frac{dy}{dt} = 0$, $\frac{dx}{dt} \neq 0$).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Horizontal and vertical tangents on parametric curves

For the same curve, find the value(s) of t where the tangent line is vertical ($\frac{dx}{dt} = 0$, $\frac{dy}{dt} \neq 0$).

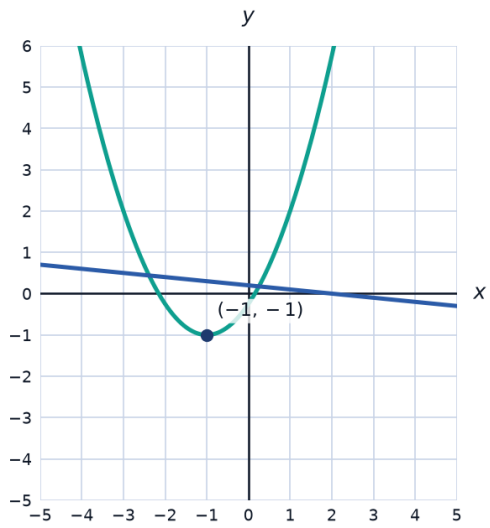
9 [Intermediate] • Target: Converting between polar and Cartesian for calculus

Find the slope of the tangent line to $r = 1 + \sin\theta$ at $\theta = 0$, using $x = r\cos\theta$, $y = r\sin\theta$ as parametric equations in θ.

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10 **[Intermediate]** • **Target: Speed and length for parametric motion**

A particle moves according to $x(t) = t^2$, $y(t) = 4t$. Find its speed at $t = 3$.



11 **[Intermediate]** • **Target: Speed and length for parametric motion**

Find the arc length of $x = \cos^3 t$, $y = \sin^3 t$ for $0 \le t \le \frac{\pi}{2}$ (an astroid arc), using $\frac{dx}{dt} = -3\cos^2 t \sin t$ and $\frac{dy}{dt} = 3\sin^2 t \cos t$.

12 **[Intermediate]** • **Target: Slope and concavity for polar-derived curves**

Find $\frac{dy}{dx}$ for the polar curve $r = 2\theta$ at $\theta = \frac{\pi}{2}$, using $x = r\cos\theta$, $y = r\sin\theta$.

13 **[Intermediate]** • **Target: Slope and concavity for polar-derived curves**

For $x = t^3 - 3t$, $y = t^2$, find the points where the tangent is horizontal.

14 **[Intermediate]** • **Target: Arc length of a polar curve**

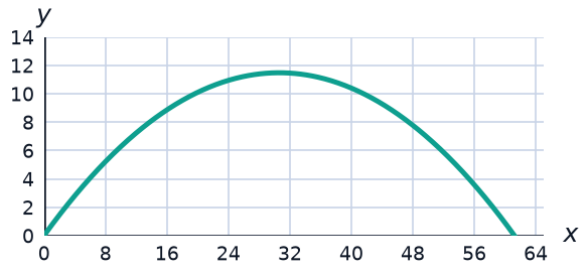
Set up (do not evaluate) the arc length integral for the circle $r = 4$ over $0 \le \theta \le 2\pi$, using $L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$, and verify the setup is consistent with the known circumference formula.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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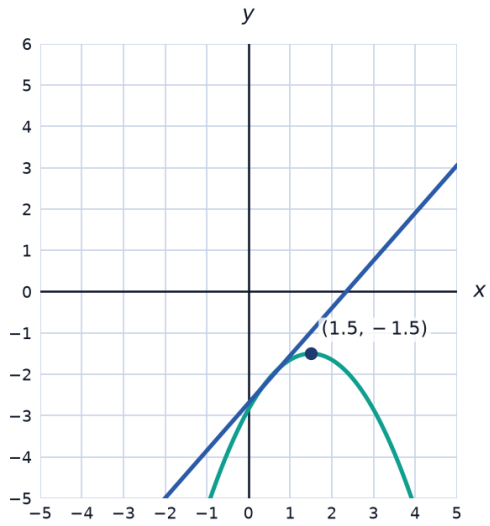
15 [Challenging] • Target: Applied parametric motion • Cross-Lesson: 6.01 Related Rates

A projectile's position is given by $x(t) = 20t$, $y(t) = 15t - 4.9t^2$ (meters, seconds). Find the projectile's velocity components $\frac{dx}{dt}$ and $\frac{dy}{dt}$, and find the speed at $t = 1$.



16 [Challenging] • Target: Applied parametric motion • Cross-Lesson: 6.02 Optimization and Curve Sketching

For the same projectile, find the time at which the vertical velocity is 0 (the peak of the trajectory).



17 [Challenging] • Target: Polar curves and symmetry • Cross-Lesson: 7.01 Integration by Substitution

The cardioid $r = 1 + \cos\theta$ is symmetric about the polar axis. Use this symmetry to explain why the total area enclosed can be computed as $2 \times$ the area for $0 \le \theta \le \pi$, and evaluate the total area.

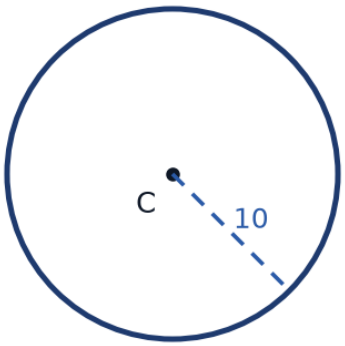
18 [Challenging] • Target: Polar curves and symmetry • Cross-Lesson: 7.02 Integration by Parts

For the parametric curve $x = e^t\cos t$, $y = e^t\sin t$, find $\frac{dy}{dx}$ at $t = 0$, using $\frac{dx}{dt} = e^t(\cos t - \sin t)$ and $\frac{dy}{dt} = e^t(\sin t + \cos t)$.

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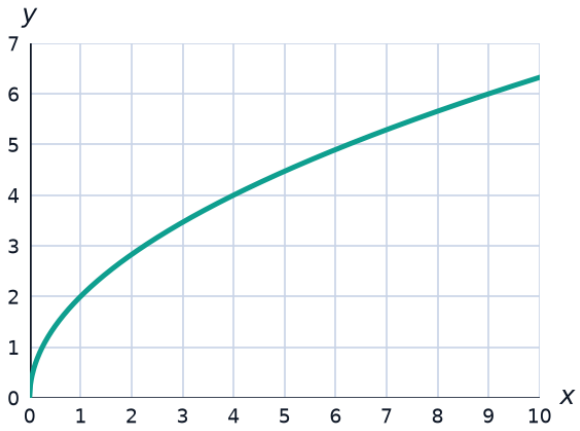
19 [Challenging] • Target: Polar curves and symmetry • Cross-Lesson: 8.01 Area Between Curves

A Ferris wheel of radius 10 m, centered 12 m above the ground, rotates so that a rider's position is $x(t) = 10\cos t$, $y(t) = 12 + 10\sin t$, where t is measured in radians and represents elapsed angle since the rider was at the rightmost point. Find the rider's height above the ground when $t = \frac{\pi}{3}$, and find the vertical velocity $\frac{dy}{dt}$ at that instant (assuming t increases at a constant rate of 1 rad per unit time).



20 [Challenging] • Target: Total distance traveled along a parametric path • Cross-Lesson: 8.02 Volumes of Revolution

A robot on a factory floor follows the path $x(t) = t^2$, $y(t) = 2t$ for $0 \le t \le 3$ (meters, seconds). Set up the arc length integral for the total distance traveled, and evaluate it exactly using the substitution $u = t^2 + 1$ where appropriate.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Parametric derivatives and arc length

Arc length of $r = a\sin^2(\theta/2)$? Recognise this equals $a(1-\cos\theta)/2$. Total length:

a. $4a$

b. $2a$

c. $8a$

d. πa

22 [Foundational] • Target: Polar area

Polar area between two curves $r_1(\theta) \le r_2(\theta)$:

a. $\frac{1}{2}\int (r_2^2 - r_1^2)d\theta$

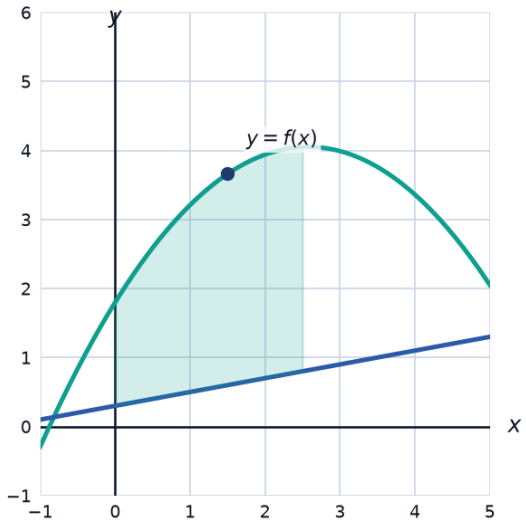
b. $\frac{1}{2}\int (r_2 - r_1)^2 d\theta$

c. $\int (r_2 - r_1)d\theta$

d. $\pi(r_2^2 - r_1^2)$

23 [Foundational] • Target: Second derivatives and concavity for parametric curves

Consider the parametric curve defined by $x(t) = 3\cos(t) - \cos(3t)$ and $y(t) = 3\sin(t) - \sin(3t)$ for $t \in [0, 2\pi]$. This curve traces out a shape known as a 'cardioid'. Find the maximum area of a triangle that can be inscribed within the loop of this cardioid, with one vertex at the origin.



a. $\frac{27\sqrt{3}}{2}$

b. $\frac{81}{2}$

c. $\frac{81\sqrt{3}}{4}$

d. $54\sqrt{3}$

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24 [Foundational] • Target: Area between polar curves

A rectangular garden is to be enclosed by a fence. One side of the garden is adjacent to a long straight river, and requires no fencing. The other three sides require fencing, and the total length of the fence available is 120 meters. Let the side of the garden parallel to the river have length x meters, and let the two sides perpendicular to the river each have length y meters. The area of the garden is given by $A(x, y) = xy$. \\ Find the dimensions x and y that maximize the area of the garden.

a. x = 60 m, y = 30 m

b. x = 30 m, y = 60 m

c. x = 40 m, y = 40 m

d. x = 80 m, y = 20 m

25 [Foundational] • Target: Horizontal and vertical tangents on parametric curves

A particle's position in the xy-plane is given by the parametric equations $x(t) = 2t + 1$ and $y(t) = t^2 - 3$, where t is time. What is the Cartesian equation of the path traced by the particle?

a. $y = \frac{1}{2}x^2 - 3$

b. $y = \left(\frac{x-1}{2}\right)^2 - 3$

c. $y = x^2 - 3$

d. $y = 2\left(\frac{x-1}{2}\right) + 1$

26 [Foundational] • Target: Converting between polar and Cartesian for calculus

Epicycloid with n cusps (outer radius R, small circle radius $r = R/n$) has arc length:

a. $8nR$

b. $8R$

c. $4nR$

d. $2\pi nR$

27 [Foundational] • Target: Speed and length for parametric motion

Arc length of the parametric curve $x = t^2$, $y = t^3$, $0 \le t \le 1$:

a. $\sqrt{13} - 1$

b. $\frac{1}{27}(13\sqrt{13} - 8)$

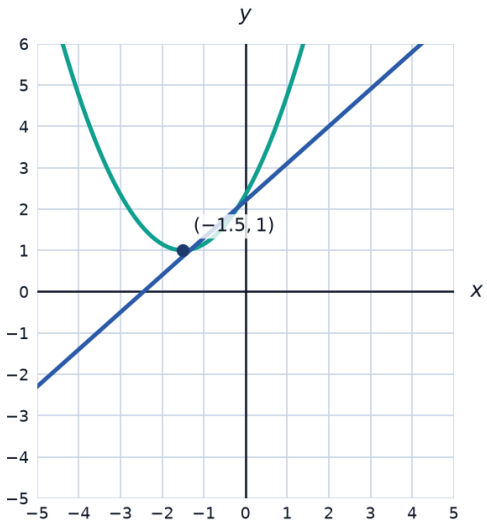
c. $\frac{2}{27}(13\sqrt{13} - 8)$

d. 1

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28 [Foundational] • Target: Slope and concavity for polar-derived curves

Find the length of the curve $x = t^2$, $y = \frac{t^3}{3} - t$, $0 \le t \le 1$.



a. $\frac{8}{3}$

b. $\frac{2}{3}$

c. $\frac{7}{3}$

d. $\frac{4}{3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Arc length of a polar curve

If $r = f(\theta)$, the polar arc length is:

a. $\int \sqrt{r^2 + (dr/d\theta)^2}\, d\theta$

b. $\int r\, d\theta$

c. $\int r^2\, d\theta$

d. $\int |dr/d\theta| d\theta$

30 [Intermediate] • Target: Applied parametric motion

Length of the curve $r = a\sec^2(\theta/2)$ (parabola) from $\theta = 0$ to $\theta = \pi/2$:

a. $a(\sqrt{2} + \ln(1 + \sqrt{2}))$

b. $a\sqrt{2}$

c. $2a$

d. $\pi a/2$

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31 [Intermediate] • Target: Polar curves and symmetry

Tangent line to parametric curve is horizontal when:

a. $dx/dt = 0$

b. $dy/dt = 0$ and $dx/dte0$

c. $t = 0$

d. $dy/dt = dx/dt$

32 [Intermediate] • Target: Total distance traveled along a parametric path

Tangent line to parametric curve is vertical when:

a. $dy/dt = 0$

b. $dx/dt = 0$ and $dy/dte0$

c. $dx/dt = dy/dt$

d. $t = 0$

33 [Intermediate] • Target: Parametric derivatives and arc length

Arc length of hypocycloid $x = a\cos^3 t, y = a\sin^3 t$ (astroid):

a. $6a$

b. $8a$

c. $4a$

d. $3a$

34 [Intermediate] • Target: Polar area

Find the total length of the cardioid $r = 2 - 2\cos(\theta)$.

a. 16

b. 8

c. 4π

d. 32

35 [Intermediate] • Target: Second derivatives and concavity for parametric curves

Find the total area enclosed by the rose $r = \sin(6\theta)$.

a. π

b. $\frac{\pi}{2}$

c. $\frac{\pi}{4}$

d. $\frac{\pi}{24}$

36 [Intermediate] • Target: Area between polar curves

Arc length of cycloid one arch: $x = r(t - \sin t), y = r(1 - \cos t), 0 \le t \le 2\pi$:

a. $4r$

b. $8r$

c. $2\pi r$

d. $6r$

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37 [Intermediate] • Target: Horizontal and vertical tangents on parametric curves

Lemniscate $r^2 = a^2\cos 2\theta$: total area enclosed:

a. a^2

b. $\pi a^2/2$

c. $2a^2$

d. πa^2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Converting between polar and Cartesian for calculus • Cross-Lesson: 7.02 Integration by Parts

The area between the outer loop and inner loop of $r = 1 + 2\cos\theta$:

a. $2\pi - 3\sqrt{3}/2$

b. π

c. $2\pi + \sqrt{3}$

d. 3π

39 [Challenging] • Target: Speed and length for parametric motion • Cross-Lesson: 8.01 Area Between Curves

Signed area of a parametric closed curve traversed counterclockwise:

a. $\int_a^b y\dot{x}\,dt$

b. $-\int_a^b y\dot{x}\,dt$

c. $\frac{1}{2}\int_a^b (x\dot{y} - y\dot{x})dt$

d. $\int_a^b x\dot{y}\,dt$

40 [Challenging] • Target: Slope and concavity for polar-derived curves • Cross-Lesson: 8.02 Volumes of Revolution

Spherical coordinates: arc length element $ds^2 =$

a. $dr^2 + d\theta^2 + d\phi^2$

b. $dr^2 + r^2 d\theta^2 + r^2\sin^2\theta\, d\phi^2$

c. $dr^2 + r^2 d\phi^2$

d. $dr^2 + r^2 d\theta^2$

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41 [Challenging] • Target: Arc length of a polar curve • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Area of region bounded by parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (full astroid):

a. $3\pi/8$

b. $3\pi/4$

c. $\pi/4$

d. $\pi/2$

42 [Challenging] • Target: Applied parametric motion • Cross-Lesson: 9.01 Separable Differential Equations

Rotating a parametric curve $\{(x(t), y(t))\}$ about the x-axis, surface area:

a. $\int 2\pi y\,|\dot{\mathbf{r}}|\,dt$

b. $\pi\int y^2\,dt$

c. $2\pi\int x\,dt$

d. $\int y\,dt$

43 [Challenging] • Target: Polar curves and symmetry • Cross-Lesson: 1.01 Precalculus Review for Calculus

Normal to parametric curve at $t = t_0$ has slope:

a. $\frac{dx/dt}{dy/dt}$

b. $-\frac{dx/dt}{dy/dt}$

c. $\frac{dy/dt}{dx/dt}$

d. $-\frac{dy/dt}{dx/dt}$

44 [Challenging] • Target: Total distance traveled along a parametric path • Cross-Lesson: 11.01 Sequences and Series

Speed of a parametric curve: $|\mathbf{v}| =$

a. $|dy/dt|$

b. $\sqrt{(dx/dt)^2 + (dy/dt)^2}$

c. $dx/dt + dy/dt$

d. $|(dx + dy)/dt|$

45 [Challenging] • Target: Parametric derivatives and arc length • Cross-Lesson: 12.01 Convergence Tests

Slope dy/dx of parametric curve:

a. $\frac{dy/dt}{dx/dt}$

b. $\frac{dx/dt}{dy/dt}$

c. $dy \cdot dx$

d. dt

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